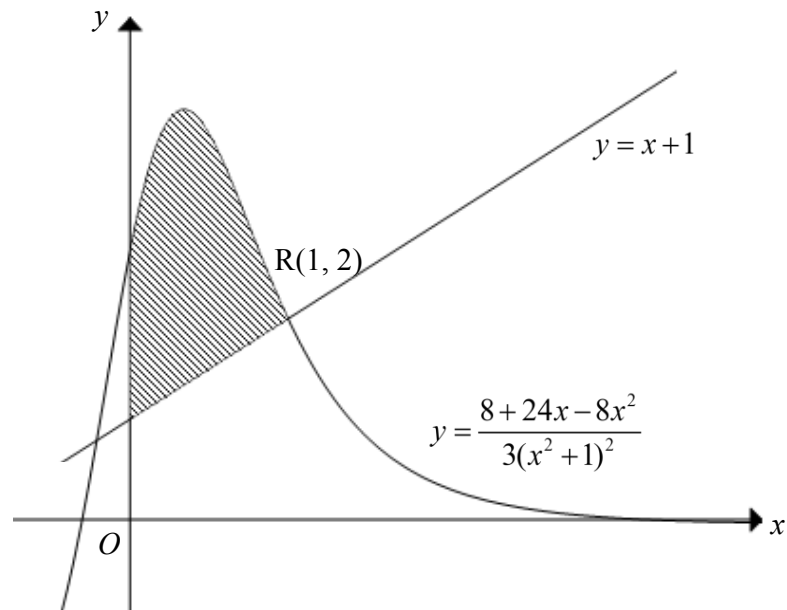


- (b) The diagram shows part of the curve $y = \frac{8+24x-8x^2}{3(x^2+1)^2}$. The line $y = x+1$ intersects the curve at $R(1, 2)$. Find the area of the shaded region. [5]



- 12.** In the expansion of $\left(x^2 - \frac{k}{x^2}\right)^8$, where k is a positive constant, the term independent of x is 5670.

(a) Show that $k = 3$.

[4]

- (b)** With this value of k , find the coefficient of the x^4 term in the expansion of $\left(x^2 - \frac{k}{x^2}\right)^8 (2 + x^4)$.

[4]